

Table-Heavy Electrical Specification Fixture

Synthetic selectable-text table with dense units, ranges, conditions, and footnotes.

Parameter	Symbol	Min	Typ	Max	Unit	Condition	Notes
Input open-circuit voltage	Voc	0	-	125	Vdc	-40 to 85 degC	absolute max
MPPT operating window	Vmppt	12.5	-	105	Vdc	Pout>50 W	tracking range
Maximum short-circuit current	Isc	-	-	20	Adc	module STC	per input
Continuous input current	Iin	-	15	18	Adc	Tamb<=65 degC	derate above 65 degC
Rated input power	Pin	-	1100	1300	W	STC module	model dependent
Output operating voltage	Vout	5	-	80	Vdc	normal operation	optimizer bus
Safety output voltage	Vsafe	0.9	1.0	1.1	Vdc	standby	per optimizer
Maximum output current	Iout	-	18	20	Adc	continuous	model dependent
Peak conversion efficiency	eta_pk	-	99.5	-	%	nominal Vin	datasheet style
Weighted efficiency	eta_w	-	98.8	-	%	CEC weighted	typical
Isolation withstand	Viso	3000	-	-	Vac	60 s hipot	production test
Leakage current	Ileak	-	0.2	0.5	mA	264 Vac equiv	dry condition
Operating temperature	Tamb	-40	-	85	degC	free air	derating applies
Storage temperature	Tstg	-40	-	85	degC	non-operating	packed unit

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Relative humidity	RH	0	-	100	%	condensing	sealed enclosure
Ingress protection	IP	-	68	-	-	IEC 60529	field installed
Connector contact resistance	Rc	-	1.0	2.5	mohm	new mated pair	per pole
Output wire length positive	Lplus	4.7	5.3	-	m	factory lead	routing limit
Output wire length negative	Lminus	0.10	0.10	-	m	factory lead	routing limit
Maximum system voltage	Vsys	-	1000	-	Vdc	array design	code dependent
Altitude	Alt	0	-	2000	m	without derating	site condition
Pollution degree	PD	-	3	-	-	outdoor	insulation coord.
Overvoltage category	OVC	-	II	-	-	DC port	IEC 62109
Fire rating material	UL94	-	V-0	-	-	polymer enclosure	flame class
Torque mounting bolt	Mbolt	8	10	12	N m	stainless rail	installer check
Minimum clearance	Clr	25	-	-	mm	rear surface	thermal airflow
Ground bond resistance	Rb	-	-	100	mohm	PE continuity	if bonded
Power-on start delay	tdly	-	35	60	s	grid present	firmware controlled